Software Testing Plan

Project Name:- The Temperature Converter

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1. Introduction

This document outlines the testing strategy, scope, and test cases for the Temperature Converter application. The objective is to ensure the conversion tool accurately converts temperatures between Celsius, Fahrenheit, and Kelvin, providing users with precise data for scientific or daily use.

2. Test Scope

- **In-Scope**: User Interface (UI) functionality, input validation (numeric values only), and conversion algorithm accuracy .

- **Out-of-Scope**: Application installation process, long-term data storage, and network speed testing.

3. Test Environment

- **Automation Framework**: Selenium WebDriver
- **Programming Languages**: Python, Java
- **Build Tool**: Maven
- **IDE**: Visual Studio Code (VS Code)
- **Target Browser**: Fire Fox

Test Scenario Execution Log (Excel Format)

ID	Model	Scenario	Input
TC-01	Converter UI	Verify Celsius to Fahrenheit	0°C
TC-02	Converter UI	Verify Fahrenheit to Celsius	212°F
TC-03	Converter UI	Verify Celsius to Kelvin	25°C
TC-04	Converter UI	Verify Invalid Input Handling	ABC

Write a Junit testing using Java.

Aim: To write a Junit testing using Java**.**

App file: Temperatureconverter.java

Program:

```
package junit;

public class TemperatureConverter
{
   public static double celsiusTofahrenheit(double celsius){
  return((celsius * 9 / 5) + 32);
   }
   public static double fahrenheitToCelsius(double fernheit){
      return((fernheit - 32) * 5/2);
   }
}
```

App file: temperatureconverterTest.java

Program:-

```
package junit;
 import static org.junit.jupiter.api.Assertions.*;
 import org.junit.jupiter.api.Test;
public class TempertureConvertortest
{
   @Test
    void testcelsiusToFahrenheit()
   {
      TemperatureConverter tem=new TemperatureConverter();
      assertEquals(212.0,tem.celsiusTofahrenheit(100.0));
   }
    void  testfahrenheittocelsius(){
      TemperatureConverter tem=new TemperatureConverter();
      assertEquals(100.0,tem. fahrenheitToCelsius(212.0));
   }
}
```

Output : (test pass)

mvn compile

```
PS C:\Users\Devendra\.vscode\junit_test>  mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< junit:junit_test >------------------------
[INFO] Building junit_test 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ junit_test ---
[INFO] skip non existing resourceDirectory C:\Users\Devendra\.vscode\junit_test\src\main\
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ junit_test ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.921 s
[INFO] Finished at: 2026-02-27T13:21:35+05:30
[INFO] ------------------------------------------------------------------------
```

mvn test

```
PS C:\Users\Devendra\.vscode\junit_test> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< junit:junit_test >------------------------
[INFO] Building junit_test 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ junit_test ---
[INFO] skip non existing resourceDirectory C:\Users\Devendra\.vscode\junit_test\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ junit_test ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.139 s
[INFO] Finished at: 2026-02-19T19:28:47+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Devendra\.vscode\junit_test> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< junit:junit_test >------------------------
[INFO] Building junit_test 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ junit_test ---
[INFO] skip non existing resourceDirectory C:\Users\Devendra\.vscode\junit_test\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ junit_test ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ junit_test ---
[INFO] skip non existing resourceDirectory C:\Users\Devendra\.vscode\junit_test\src\test\resources
```

Conclusion: The program build was Successful

If test fail :

App Test File Name : TemperatureconverterTest.Java

Program :

```
 package junit;

import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;

public class TempertureConvertorTest {

    @Test
    void testCelsiusToFahrenheit() {
        TemperatureConverter tem = new TemperatureConverter();

        assertEquals(0, tem.celsiusTofahrenheit(100.0));
    }
    @Test
    void testfahrenheittocelsius() {
        TemperatureConverter tem = new TemperatureConverter();
        // This WILL fail because the math won't match 1045
        assertEquals(1045, tem.fahrenheitToCelsius(234.0));
    }
}
```

Output : (test fail)

```
PS C:\Users\Devendra\.vscode\junit_test> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------------< junit:junit_test >-------------------------
[INFO] Building junit_test 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ junit_test ---
[INFO] skip non existing resourceDirectory C:\Users\Devendra\.vscode\junit_test\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ junit_test ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.103 s
[INFO] Finished at: 2026-03-05T19:28:45+05:30
[INFO] ------------------------------------------------------------------------
```

```
[INFO] ------------------------------------------------------------------------
PS C:\Users\Devendra\.vscode\junit_test> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------------< junit:junit_test >-------------------------
[INFO] Building junit_test 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ junit_test ---
[INFO] skip non existing resourceDirectory C:\Users\Devendra\.vscode\junit_test\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ junit_test ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ junit_test ---
[INFO] skip non existing resourceDirectory C:\Users\Devendra\.vscode\junit_test\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ junit_test ---
[INFO] Recompiling the module because of added or removed source files.
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ junit_test ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running junit.TempertureConvertorTest
[ERROR] Tests run: 2, Failures: 2, Errors: 0, Skipped: 0, Time elapsed: 0.124 s <<< FAILURE! -- in junit.TempertureConvertorTest
[ERROR] junit.TempertureConvertorTest.testfahrenheittocelsius -- Time elapsed: 0.066 s <<< FAILURE!
org.opentest4j.AssertionFailedError: expected: <1045.0> but was: <505.0>
        at org.junit.jupiter.api.AssertionFailureBuilder.build(AssertionFailureBuilder.java:151)
        at org.junit.jupiter.api.AssertionFailureBuilder.buildAndThrow(AssertionFailureBuilder.java:132)
        at org.junit.jupiter.api.AssertEquals.failNotEqual(AssertEquals.java:197)
        at org.junit.jupiter.api.AssertEquals.assertEquals(AssertEquals.java:70)
        at org.junit.jupiter.api.AssertEquals.assertEquals(AssertEquals.java:65)
        at org.junit.jupiter.api.Assertions.assertEquals(Assertions.java:885)
        at junit.TempertureConvertorTest.testfahrenheittocelsius(TempertureConvertorTest.java:18)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[ERROR] junit.TempertureConvertorTest.testCelsiusToFahrenheit -- Time elapsed: 0.084 s <<< FAILURE!
org.opentest4j.AssertionFailedError: expected: <0.0> but was: <212.0>
        at org.junit.jupiter.api.AssertionFailureBuilder.build(AssertionFailureBuilder.java:151)
        at org.junit.jupiter.api.AssertionFailureBuilder.buildAndThrow(AssertionFailureBuilder.java:132)
        at org.junit.jupiter.api.AssertEquals.failNotEqual(AssertEquals.java:197)
        at org.junit.jupiter.api.AssertEquals.assertEquals(AssertEquals.java:70)
        at org.junit.jupiter.api.AssertEquals.assertEquals(AssertEquals.java:65)
        at org.junit.jupiter.api.Assertions.assertEquals(Assertions.java:885)
        at junit.TempertureConvertorTest.testCelsiusToFahrenheit(TempertureConvertorTest.java:12)
        at java.base/java.lang.reflect.Method.invoke(Method.java:569)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1511)

[INFO]
[INFO] Results:
[INFO]
[ERROR] Failures:
[ERROR]   TempertureConvertorTest.testCelsiusToFahrenheit:12 expected: <0.0> but was: <212.0>
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-surefire-plugin:3.2.5:test (default-test) on project junit_test: There are test failures.
[ERROR]
[ERROR] Please refer to C:\Users\Devendra\.vscode\junit_test\target\surefire-reports for the individual test results.
[ERROR] Please refer to dump files (if any exist) [date].dump, [date]-jvmRun[N].dump and [date].dumpstream.
[ERROR] -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException
PS C:\Users\Devendra\.vscode\junit_test>
```

Conclusion: The program build was failed

Write a seleium testing in python

Aim: To write a seleium testing in python

App file : main.py (test pass)

Program :

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import os

# 1. Initialize Firefox
driver = webdriver.Firefox()

try:
    # 2. Open your local file (Ensure the path is correct)

    file_path = "file://" + os.path.abspath("index.html")
    driver.get(file_path)

    # 3. Perform Conversion
    driver.find_element(By.ID, "tempInput").send_keys("100")
    driver.find_element(By.TAG_NAME, "button").click()

    # 4. Logic for Pass/Fail
    result_text = driver.find_element(By.ID, "result").text

    # CHANGE THIS VALUE TO TEST PASS/FAIL
    expected_value = "100°C = 212.00°F"

    if result_text == expected_value:
        print("TEST PASS: Result matched expected value.")
    else:
        print(f"TEST FAIL: Expected '{expected_value}' but got '{result_text}'")

finally:
    driver.quit()
```

HTML File : index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Temperature Converter</title>
  <style>
    body { font-family: sans-serif; background-color: #eef2f5; display: flex; justify-content: center; align-items: center; height: 100vh; margin: 0; }
    .converter-card { background: white; padding: 30px; border-radius: 10px; box-shadow: 0 4px 8px rgba(0,0,0,0.1); width: 300px; text-align: center; }
    input[type="number"] { width: 100%; padding: 10px; margin: 10px 0; border: 1px solid #ddd; border-radius: 5px; box-sizing: border-box; }
    select { width: 100%; padding: 10px; border-radius: 5px; }
    button { background-color: #007bff; color: white; border: none; padding: 10px 20px; border-radius: 5px; cursor: pointer; margin-top: 15px; width: 100%; }
    button:hover { background-color: #0056b3; }
    #result { margin-top: 20px; font-weight: bold; font-size: 1.2em; color: #333; }
  </style>
</head>
<body>

<div class="converter-card">
  <h2>Converter</h2>
```

```
    <input type="number" id="tempInput" placeholder="Enter temperature">
    <select id="conversionType">
        <option value="cToF">Celsius to Fahrenheit</option>
        <option value="fToC">Fahrenheit to Celsius</option>
    </select>
    <button onclick="convert()">Convert</button>
    <div id="result"></div>
</div>

<script>
    function convert() {
        let inputTemp = parseFloat(document.getElementById('tempInput').value);
        let type = document.getElementById('conversionType').value;
        let resultDiv = document.getElementById('result');
        let convertedTemp;

        if (isNaN(inputTemp)) {
            resultDiv.innerHTML = "Please enter a number";
            return;
        }
        if (type === 'cToF') {
            convertedTemp = (inputTemp * 9/5) + 32;
            resultDiv.innerHTML = `${inputTemp}°C =
${convertedTemp.toFixed(2)}°F`;
        } else {
```

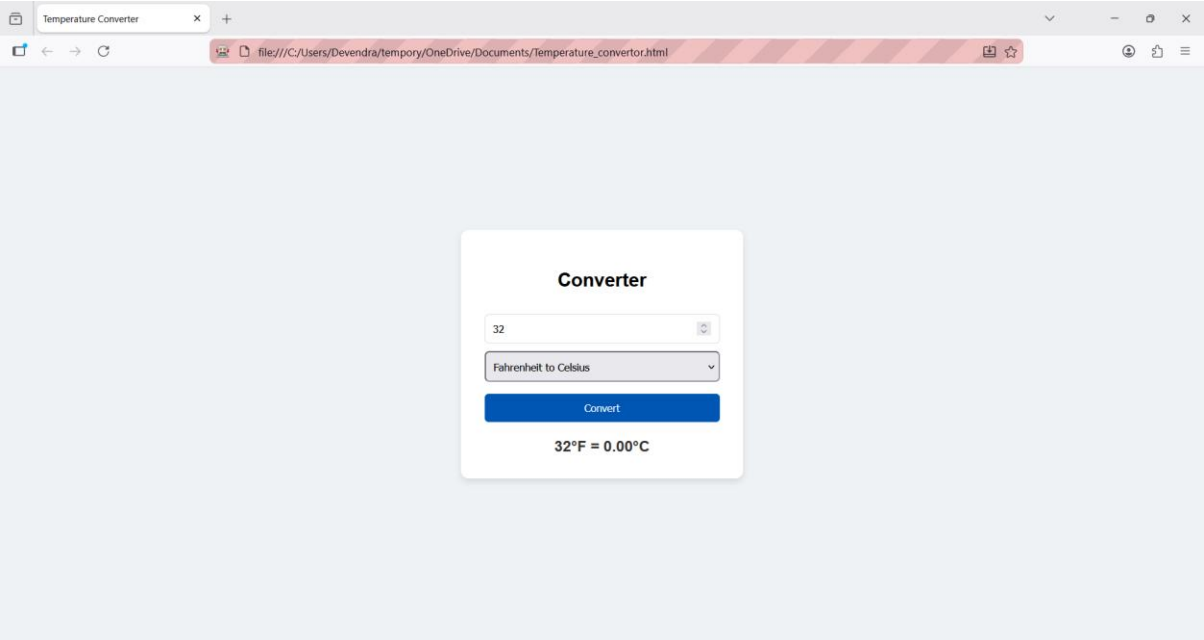
```
        convertedTemp = (inputTemp - 32) * 5/9;
        resultDiv.innerHTML = `${inputTemp}°F = ${convertedTemp.toFixed(2)}°C`;
    }
  }
</script>
</body>
</html>
```

Output : (test pass)

Command to run :Python main.py

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS        python3.11

PS C:\Selenium>  python main.py
TEST FAIL: Expected '100°C = 234.00°F' but got '100°C = 212.00°F'
```

Web page



Conclusion: The program build was Successful

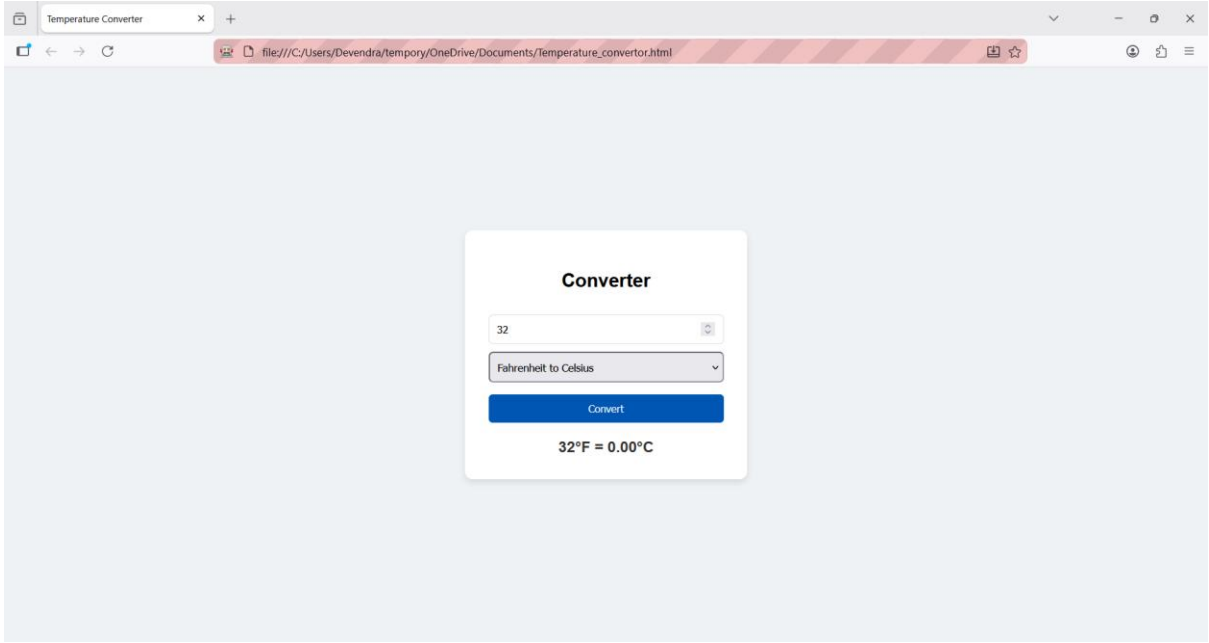
App file: main.py (test fail)

Program :

```
from selenium import webdriver
from selenium.webdriver.common.by import By
import os

# 1. Initialize Firefox
driver = webdriver.Firefox()

try:
    # 2. Open your local file (Ensure the path is correct)

    file_path = "file://" + os.path.abspath("index.html")
    driver.get(file_path)

    # 3. Perform Conversion
    driver.find_element(By.ID, "tempInput").send_keys("100")
    driver.find_element(By.TAG_NAME, "button").click()

    # 4. Logic for Pass/Fail
    result_text = driver.find_element(By.ID, "result").text

    # CHANGE THIS VALUE TO TEST PASS/FAIL
    expected_value = "100°C = 223.00°F"

    if result_text == expected_value:
        print("TEST PASS: Result matched expected value.")
    else:
        print(f"TEST FAIL: Expected '{expected_value}' but got '{result_text}'")

finally:
    driver.quit()
```

Output : (test fail)

```
PS C:\Selenium>  python main.py
TEST FAIL: Expected '100°C = 223.00°F' but got '100°C = 212.00°F'
```



Conclusion: The program build was failed

Write a seleium testing in Java.

App file : SeleniumTest.java (test pass)

Program :

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import java.io.File;

public class SeleniumTest{
   public static void main(String[] args) {
      // 1. Initialize Firefox
      WebDriver driver = new FirefoxDriver();

      try {
         // 2. Open local file
         File file = new File("index.html");
         driver.get(file.getAbsolutePath());

         // 3. Perform Conversion
         driver.findElement(By.id("tempInput")).sendKeys("0");
         driver.findElement(By.tagName("button")).click();

         // 4. Logic for Pass/Fail
         String resultText = driver.findElement(By.id("result")).getText();

         // CHANGE THIS VALUE TO TEST PASS/FAIL
         String expectedValue = "0°C = 32.00°F"; // Note: Default is C to F, this
might fail!

         if (resultText.equals(expectedValue)) {
            System.out.println("TEST PASS: Result matched expected value.");
         } else {
            System.out.println("TEST FAIL: Expected [" + expectedValue + "] but
got [" + resultText + "]");
```

```
        }
    } finally {
        driver.quit();
    }
  }
}
```

App file: TemperatureConverter.java

Program:

```
public class TemperatureConverter
{
   public static double celsiusTofahrenheit(double celsius){
 return((celsius * 9 / 5) + 32);
   }
   public static double fahrenheitToCelsius(double fernheit){
      return((fernheit - 32) * 5/2);
   }

}
```

HTML File : Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Temperature Converter</title>
   <style>
      body { font-family: sans-serif; background-color: #eef2f5; display: flex;
justify-content: center; align-items: center; height: 100vh; margin: 0; }
      .converter-card { background: white; padding: 30px; border-radius: 10px;
box-shadow: 0 4px 8px rgba(0,0,0,0.1); width: 300px; text-align: center; }
      input[type="number"] { width: 100%; padding: 10px; margin: 10px 0;
border: 1px solid #ddd; border-radius: 5px; box-sizing: border-box; }
      select { width: 100%; padding: 10px; border-radius: 5px; }
      button { background-color: #007bff; color: white; border: none; padding:
10px 20px; border-radius: 5px; cursor: pointer; margin-top: 15px; width: 100%;
}
      button:hover { background-color: #0056b3; }
      #result { margin-top: 20px; font-weight: bold; font-size: 1.2em; color: #333;
}
   </style>
</head>
<body>

<div class="converter-card">
   <h2>Converter</h2>
   <input type="number" id="tempInput" placeholder="Enter temperature">
   <select id="conversionType">
      <option value="cToF">Celsius to Fahrenheit</option>
      <option value="fToC">Fahrenheit to Celsius</option>
   </select>
   <button onclick="convert()">Convert</button>
   <div id="result"></div>
</div>
```

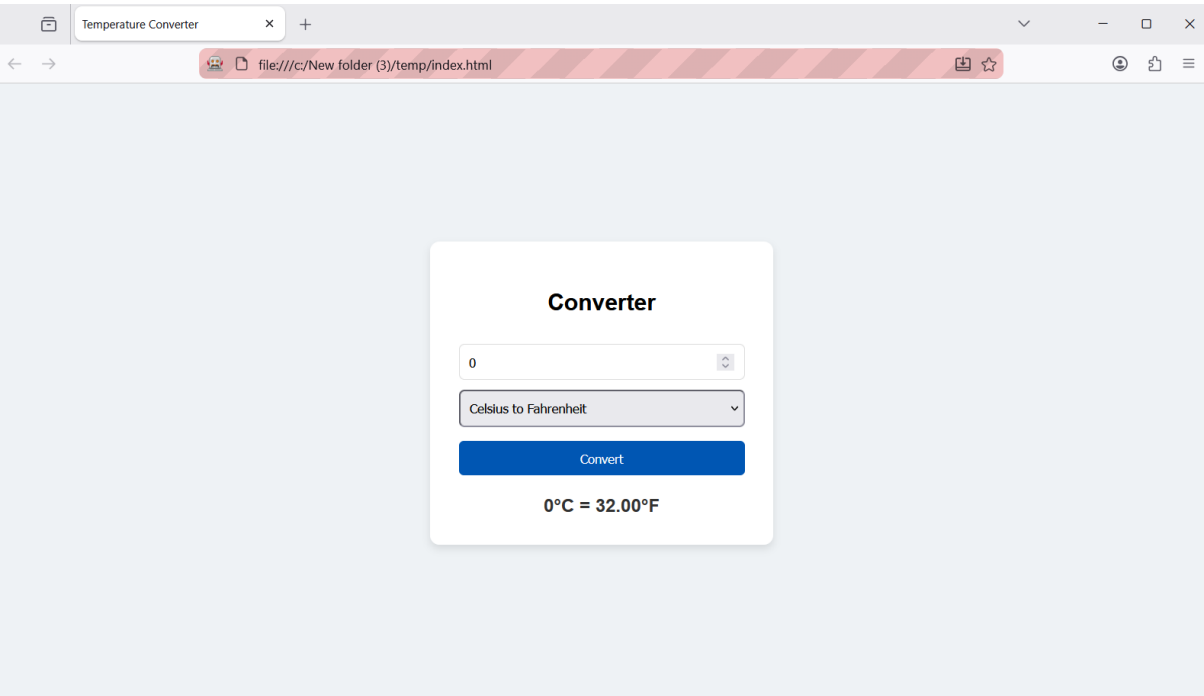
```
<script>
   function convert() {
      let inputTemp = parseFloat(document.getElementById('tempInput').value);
      let type = document.getElementById('conversionType').value;
      let resultDiv = document.getElementById('result');
      let convertedTemp;

      if (isNaN(inputTemp)) {
         resultDiv.innerHTML = "Please enter a number";
         return;
      }

      if (type === 'cToF') {
         convertedTemp = (inputTemp * 9/5) + 32;
         resultDiv.innerHTML = `${inputTemp}°C =
${convertedTemp.toFixed(2)}°F`;
      } else {
         convertedTemp = (inputTemp - 32) * 5/9;
         resultDiv.innerHTML = `${inputTemp}°F =
${convertedTemp.toFixed(2)}°C`;
      }
   }
</script>

</body>
</html>
```

Output : (test pass)

Command to run the file

```
PS C:\New folder (3)\temp> javac -cp "lib/*;src" src/SeleniumTest.java src/TemperatureConverter.java
PS C:\New folder (3)\temp> java -cp "lib/*;src" SeleniumTest
TEST PASS: Result matched expected value.
```

Web page



Conclusion : The program build was Successful

App file: SeleniumTest.java (test fail)

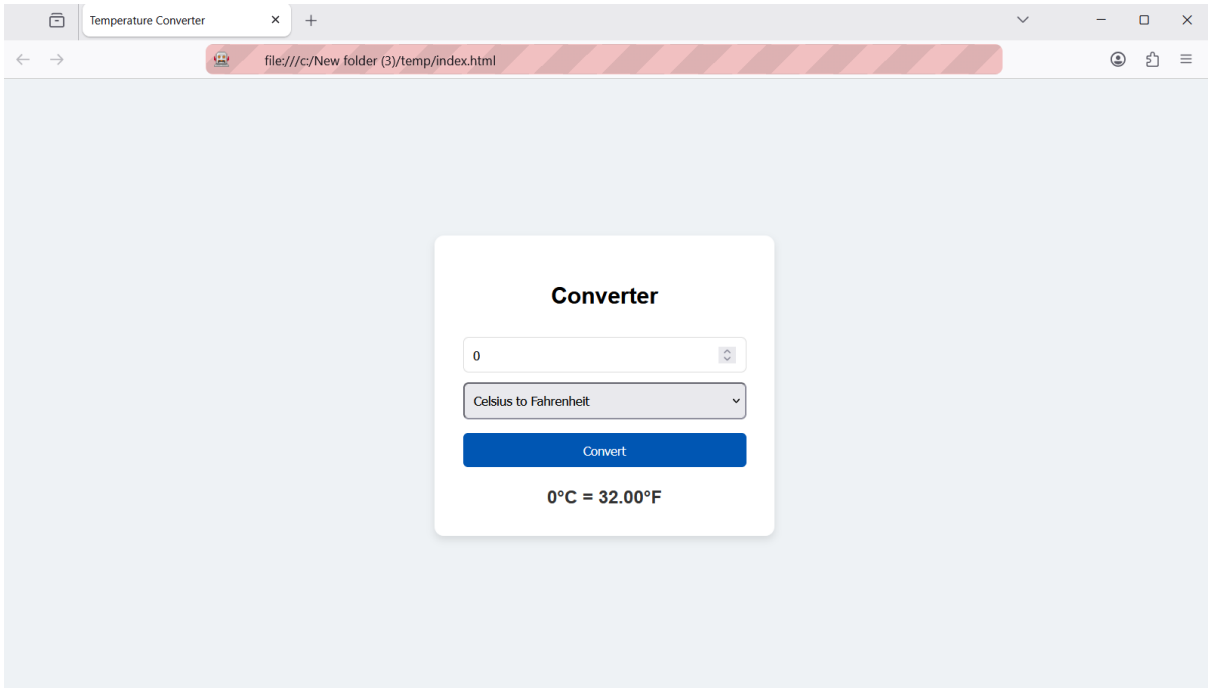
Program:

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import java.io.File;

public class SeleniumTest{
   public static void main(String[] args) {

      WebDriver driver = new FirefoxDriver();

      try {

         File file = new File("index.html");
         driver.get(file.getAbsolutePath());


         driver.findElement(By.id("tempInput")).sendKeys("0");
         driver.findElement(By.tagName("button")).click();


         String resultText = driver.findElement(By.id("result")).getText();


         String expectedValue = "0°F = 32.00°C"; // Note: Default is C to F, this
might fail!

         if (resultText.equals(expectedValue)) {
            System.out.println("TEST PASS: Result matched expected value.");
         } else {
            System.out.println("TEST FAIL: Expected [" + expectedValue + "] but
got [" + resultText + "]");
         }
```

```
        } finally {
            driver.quit();
        }
    }
}
```

Output : (test fail)

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                    powershell

PS C:\New folder (3)\temp> javac -cp "lib/*;src" src/SeleniumTest.java src/TemperatureConverter.java
PS C:\New folder (3)\temp> java -cp "lib/*;src" SeleniumTest
TEST FAIL: Expected [0°F = -17.78°C] but got [0°C = 32.00°F]
PS C:\New folder (3)\temp>
```



Conclusion : The program build was failed

1.Unit Testing Environment (JUnit)

This environment is used for testing the core logic of the application.

- Language: Java
- Build Tool: Maven
- Core File:Temperatureconvertor.java, TemperatureconvertorTest.java
- Dependency Management: pom.xml
- IDE: Visual Studio Code

2. Web Automation Environment (Selenium Java)

Used for automated UI testing of web components.

- Languages: Java, HTML
- Framework: Selenium WebDriver
- App Files: SeleniumTest.java, TemperatureConverter.java,index.html
- Target Browser: Mozilla Firefox
- IDE: Visual Studio Code

3. Web Automation Environment (Selenium Python)

A parallel environment for cross-language automation testing.

- Languages: Python, HTML
- Framework: Selenium WebDriver
- App Files: main.py, index.html
- Target Browser: Mozilla Firefox
- IDE: Visual Studio Code